

Tokenomics



Tauhid Zaman

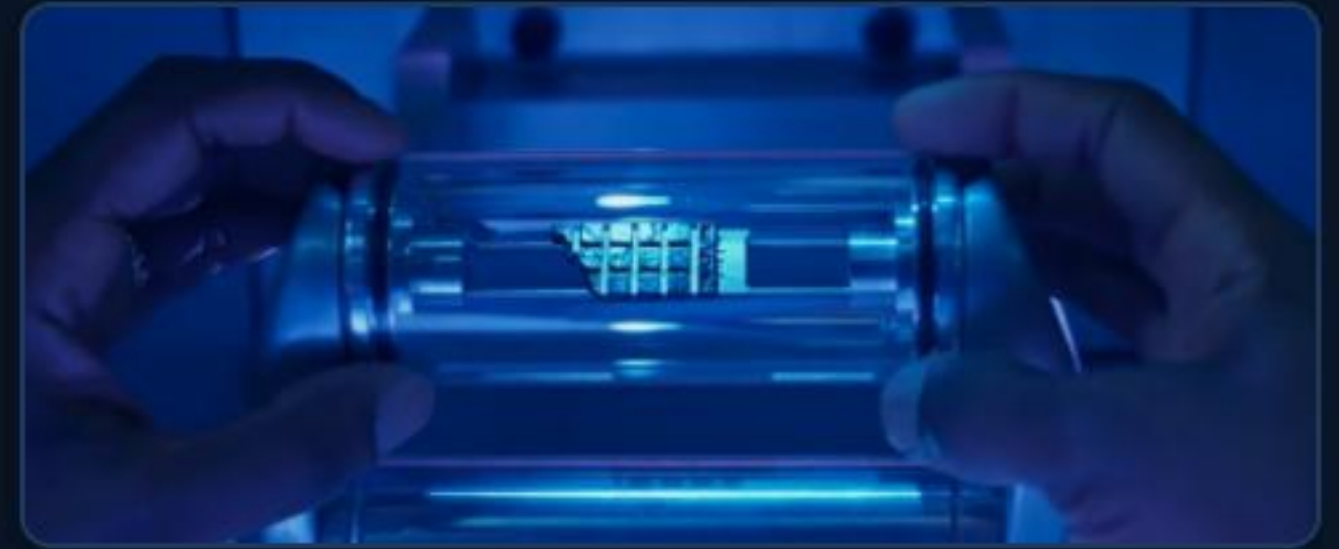
Why AI Costs What It Costs

- Tokenomics — the economics behind AI token prices
- **Hardware** — GPUs and memory: what the chips cost
- **Power** — the electricity used by the chips
- **Environment** — impact of AI datacenters on power, water, and living conditions



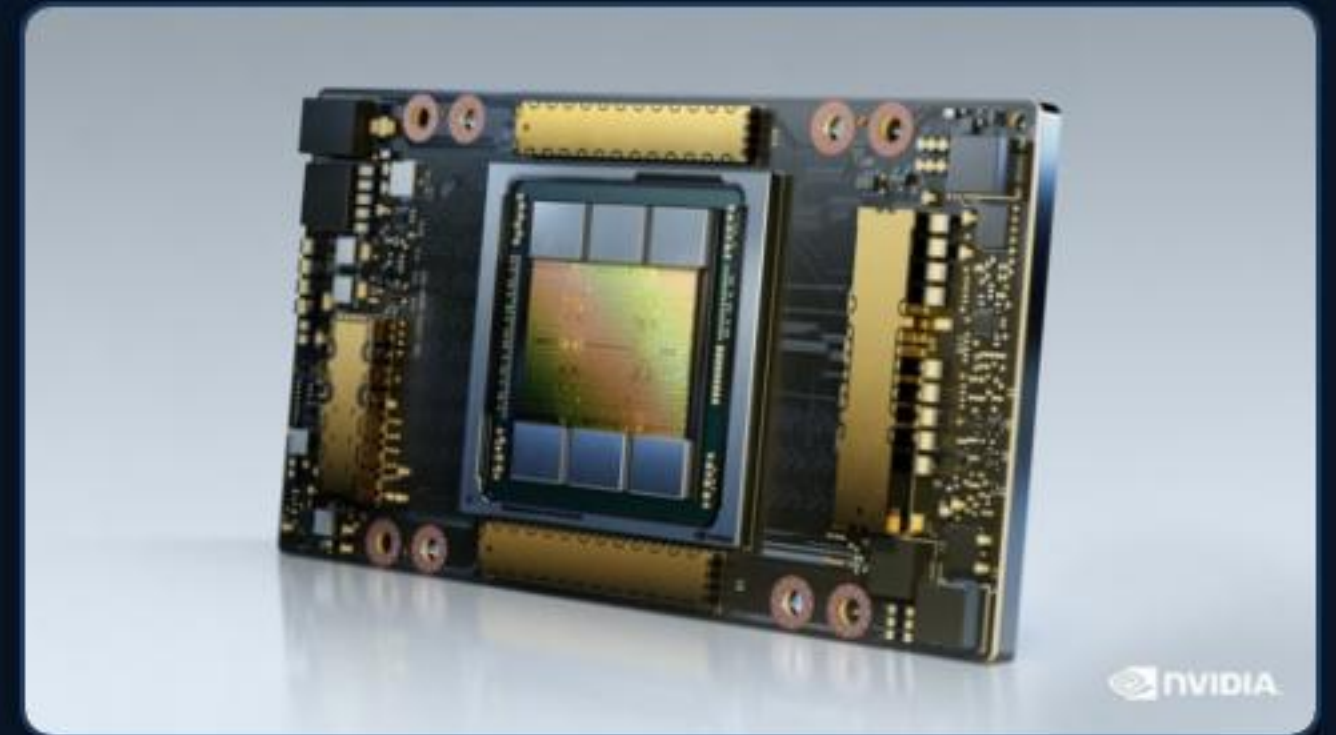
| GPUs - The Chips That Power AI

- **GPU** = graphics processing unit
- Computer chips with thousands of small cores that do the *same math* in parallel
- Main uses:
 - Computer graphics (i.e. gaming)
 - Blockchain (i.e. mining crypto)
 - AI (i.e. training and token generation)
- Fast matrix multiplication — the main math operation in AI models



| GPUs - The Chips That Power AI

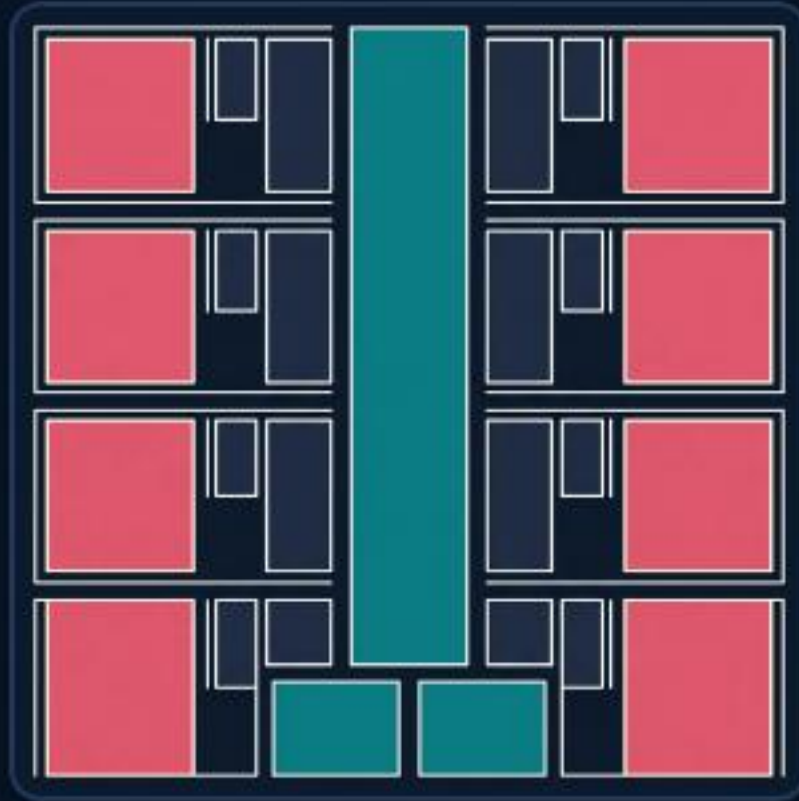
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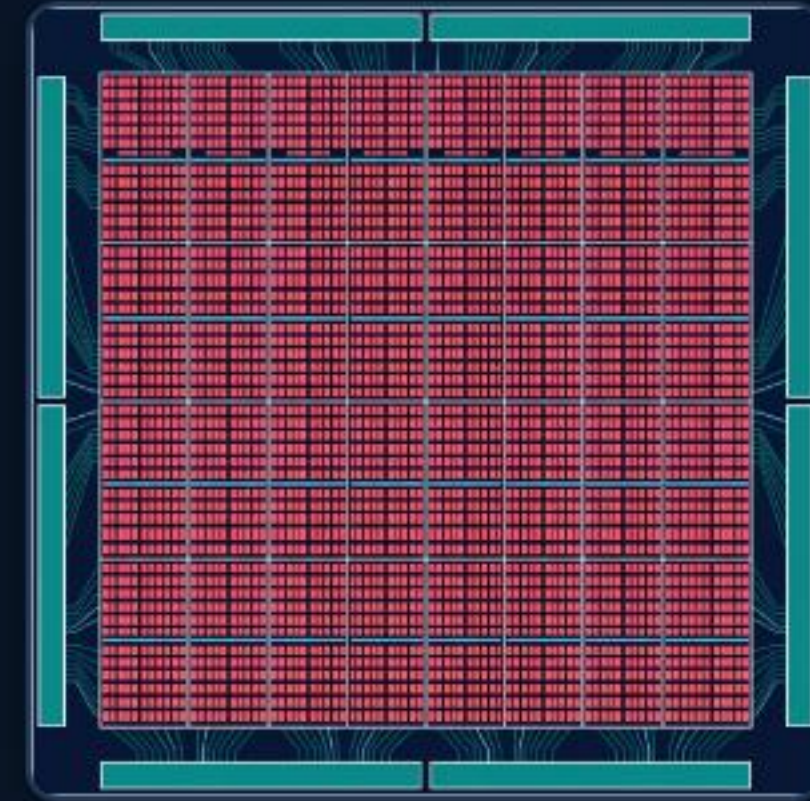
CPU vs GPU: Architecture

- **CPU** (central processing unit) — a few powerful cores; great at complex, branching, one-at-a-time logic
- **GPU** (graphics processing unit) — thousands of tiny cores; great at parallel simple computations
- AI model computations are built for thousands of tiny processors

CPU — few big cores



GPU — many tiny cores



CPU vs GPU: Metrics

AI Compute & Hardware Economics

150× compute

CPU server

Intel Xeon 6747P specs

~\$5.4k

AMD EPYC 9575F specs

~\$5.0k

GPU accelerator

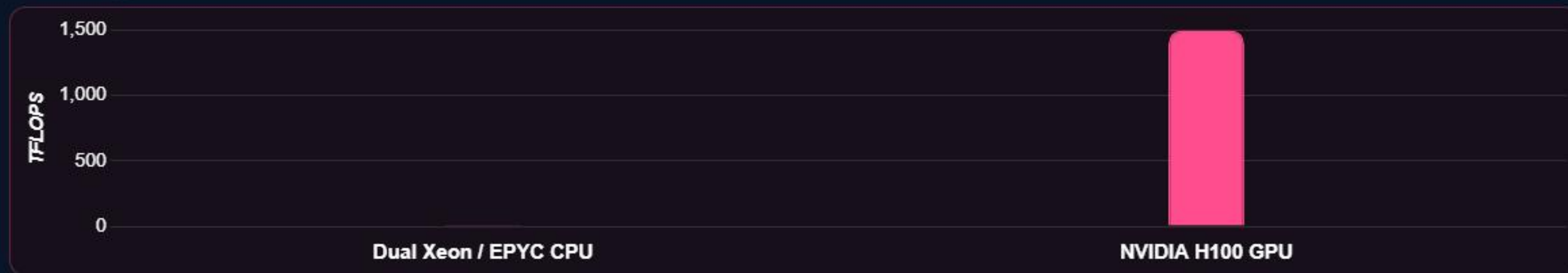
H100 specs

~\$32k

B200 specs

~\$30k–\$40k

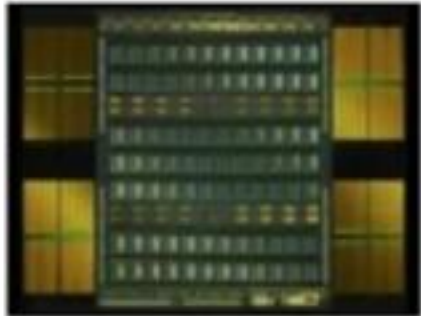
Metric **Peak throughput (TFLOPS)**



- **Peak throughput** — GPU is about 150× higher than CPU
- **Memory bandwidth** — GPU is about 11× higher than CPU
- **Cost per 1M tokens** — GPU is about 60× lower than CPU

Scaling Up GPUs

- Newer GPUs pack more transistors into the chip
 - More transistors = more computational power
- **AI accelerator** - chips built to speed up AI computation
 - GPUs are the most common accelerator
 - Other AI accelerators - TPUs (Google), Trainium (Amazon)



Volta

>21 billion transistors
815mm²
TSMC 12nm FFN

2017

Volta / V100

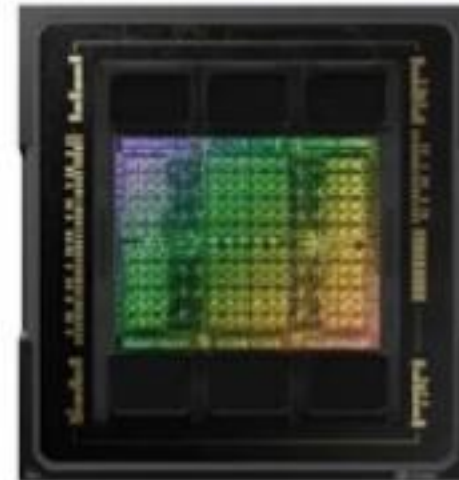


Ampere

>54 billion transistors
826 mm²
TSMC N7

2020

Ampere / A100

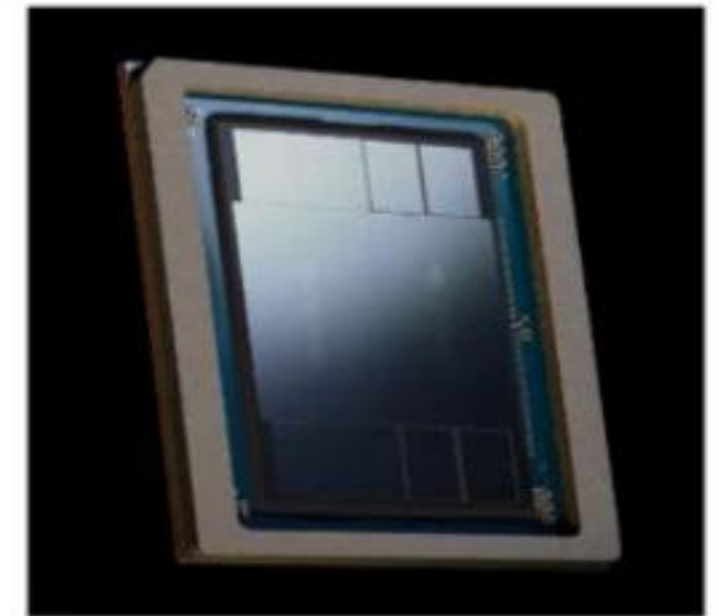


Hopper

>80 billion transistors
814 mm²
TSMC 4N

2022

Hopper / H100



Blackwell

>208 billion transistors
>1600 mm²
TSMC 4NP

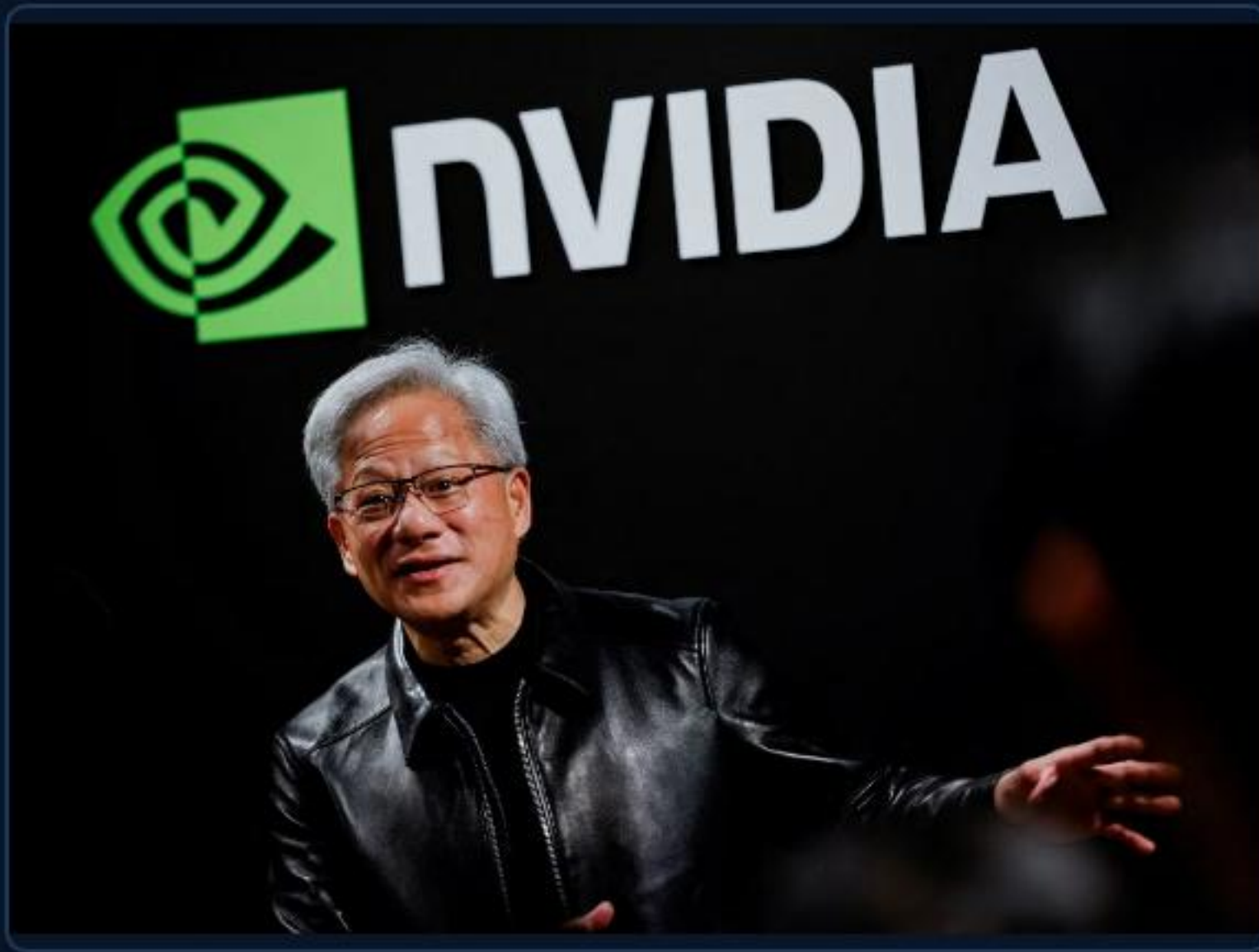
2024

Blackwell / B200

GPU Supply Chain

- **NVIDIA** designs most frontier AI GPUs — roughly ~80–90% of AI accelerator revenue
- **TSMC** (Taiwan Semiconductor Manufacturing Company) fabricates nearly **all** of those AI GPUs
- The GPU supply chain is very **concentrated** — a few firms design the chips, and one foundry builds most of them

NVIDIA — designs the chips



TSMC — builds the chips



GPU Cost

- Street prices have climbed hard each generation
- V100 era: low thousands → H100 / B200: **tens of thousands**
- AI demand + scarce supply keep them expensive

GENERATION	CHIP	PRICE
<u>Volta · 2017</u>	V100	\$1k–\$2.5k
<u>Ampere · 2020</u>	A100	\$10k–\$15k
<u>Hopper · 2022</u>	H100	\$25k–\$34k
<u>Blackwell · 2024</u>	B200	\$30k–\$55k



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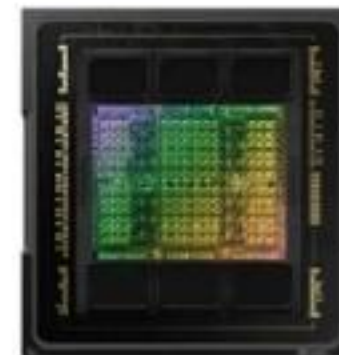


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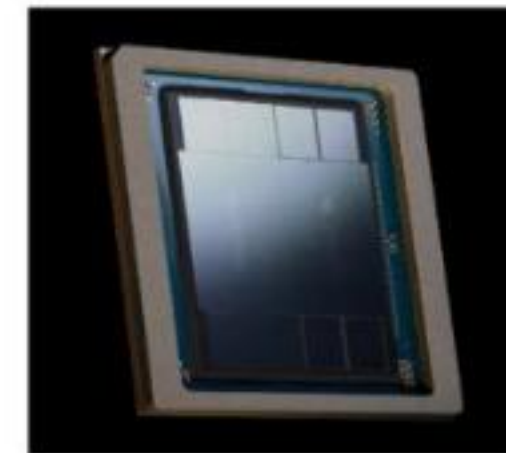


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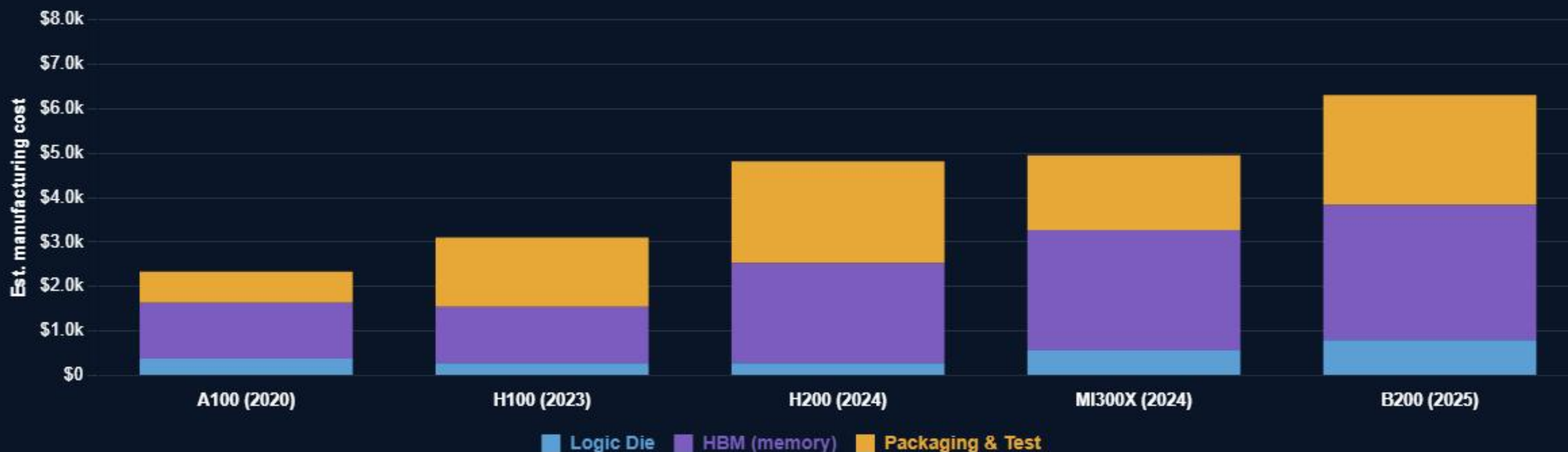
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GPU High Bandwidth Memory (HBM)

- Memory costs have increased for chips
 - GPUs use more memory
 - Memory price itself has increased
- Roughly **half** the GPU cost
- Who makes high-bandwidth memory (HBM) — Q2 2026 revenue share:
 - **SK hynix** (`000660.KS`) ~**50%**
 - **Samsung** (`005930.KS`) ~**33%**
 - **Micron** (`MU`) ~**18%**



Build-cost chart: [Silicon Analysts / Epoch.ai](#) · HBM share: [Counterpoint Q2 2026](#)

AI Accelerator Stocks

- **GPUs**

- NVIDIA (**NVDA**)
- Advanced Micro Devices (**AMD**)

- **Chip foundry**

- Taiwan Semiconductor Manufacturing Company (**TSM**)

- **Chip machines**

- ASML Holding (**ASML**) — lithography tools

- **High-bandwidth memory**

- SK hynix (**000660.KS**)
- Samsung Electronics (**005930.KS**)
- Micron (**MU**)



Cumulative return from 2018. Source: [Yahoo Finance](#) adjusted close.

How Much Power Does A GPU Use?

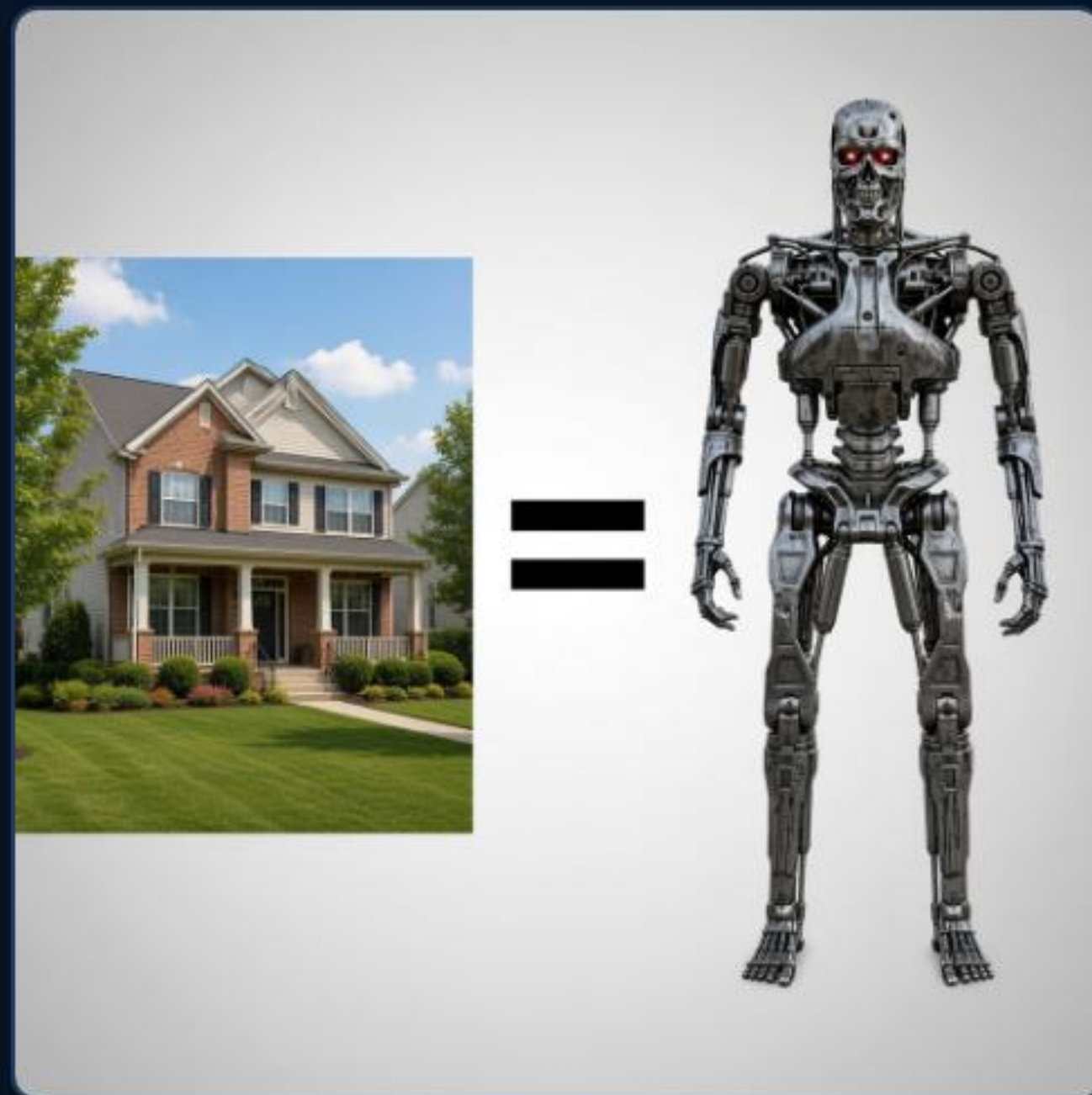
- **Chip only** — H100 up to **~700 W**; B200 **~1,000 W** ([NVIDIA](#))
- **Chip + rack share** — often **~1.3–1.7 kW** per accelerator ([SemiAnalysis](#))
- **What's in "rack power"?**
 - Power supplies (conversion losses)
 - CPUs, fans, networking cards in the server
 - Facility cooling share (chillers / liquid loops — outside the box)
 - Switches / cables that connect the GPUs
- AI datacenters are buildings full of **racks of GPU servers**
 - Photo: one **server** — **8× NVIDIA H200** — **≈ \$361k** ([Broadberry](#))

SemiAnalysis cluster power



| One GPU $\approx$ One Household

- Avg U.S. household electricity consumption $\approx$ **1.2 kW** ([EIA](#))
- Modern AI GPU power consumption $\approx$ **1 kW** ([NVIDIA H100](#))
- **1 GPU $\approx$ 1 household**
- Datacenters have thousands of GPUs



AI Datacenter Scale

- **Hyperscaler** — a company that runs huge datacenters worldwide and sells cloud compute
 - Amazon (AWS)
 - Microsoft (Azure)
 - Google (GCP)
- **Meta** — Llama 4 trained on **>100,000 H100 GPUs** [\(earnings\)](#)
- **SpaceX Colossus** (Memphis, TN) — **200,000 H100 GPUs** [\(xAI\)](#)
- 100,000 H100 GPUs $\approx$ **150 MW**
 - Big datacenters pay \$0.05–\$0.10 per kWh (homes often \$0.25–\$0.35)
 - 150 MW costs a datacenter **~\$124M/year** [\(SemiAnalysis\)](#)
- **OpenAI Stargate** (Abilene, TX) — campus planned for **~1.2 GW** [\(DCD\)](#)



100 MW = 100,000 homes of electricity use

| Water For Cooling

- GPUs get hot, so datacenters use water to cool them
- 100 MW data center uses about **500,000 gallons/day** (~2 million liters/day) ([Bloomberg / IEA](#))
- U.S. home uses about **300 gallons/day** ([EPA](#))
- 100 MW datacenter water use is equivalent to **~1,500–2,000 homes**
- **Google The Dalles** (Oregon data-center campus) — in 2024 used **~1.2M gal/day** ([reporting](#)).
 - Equivalent to 4,000 homes water use



Training AI With GPUs

- Example: **Llama 3 405B** was trained with **16,384 H100s** running for **54 days** ([Meta](#)).
- **Chip cost:** 16k chips × \$30k per chip ≈ **\$480M**
- **Power usage** (chip + rack ~1.5 kW) — 16k × 1.5 × 24 × 54 ≈ **32M kWh**
- **Power cost** — 32M kWh × \$0.10/kWh ≈ **\$3M**
- Chips dominate; electricity is still millions. Other frontier runs (many are estimates; power at ~1.5 kW/GPU):

MODEL	GPUS	TIME (DAYS)	POWER (KWH)	COST
Llama 3 405B	16k H100	~54	~32M	~\$480M chips
GPT-4	~25k A100	~95	~68M	~\$40M run
Grok-3	~100k H100	~80	~290M	~\$3B chips
Grok-4	—	—	~310M	~\$490M run

Meta · [Epoch GPT-4](#) · [Epoch Grok-4](#) · Llama/Grok-3 Cost ≈ chip CapEx · GPT-4 / Grok-4 Cost ≈ amortized training-run (Epoch)

Opposition To Datacenters

- Residents' issues
 - Turbine **noise**
 - **Water** use
 - High **electricity bills**
- Once built, few permanent **jobs** relative to the land and power footprint
- Fear: homes subsidize grid upgrades built for AI load
- Gallup (2026): **7 in 10** Americans oppose an AI datacenter nearby ([CNBC / Gallup](#))
- **Memphis / Colossus** — neighbors living with those effects
 - NAACP sued over gas turbines under the Clean Air Act ([CNBC](#))



Local Action

- **Connecticut** — New Haven + towns pause or limit builds ([NH Independent](#)) • ([CT Insider](#))
- **New Jersey** — 37 towns ban datacenters ([NJ.com](#))
- **Texas** — San Marcos citywide ban ([KUT](#))
- **Missouri** — Peculiar zoning block; St. Charles pause ([KCUR](#))
- **Oregon / Washington** — Hillsboro and Seattle pauses ([OPB](#)) • ([Seattle](#))



State Action

- **New York** — Gov. Hochul: ~1-year pause on new hyperscale (≥ 50 MW) ([NY.gov](https://www.ny.gov))
 - First statewide hyperscale pause in the U.S.; study grid, water, bills
- **New Jersey** — “Fair Share” law: datacenters pay their own power / grid costs ([NJ.gov](https://www.nj.gov))
- **Virginia** — law lets regulators shift more energy costs onto datacenters (SB 253) ([LegiScan](https://legiscan.com))



Federal Action

- **AI Data Center Moratorium Act** — Sen. Sanders + Rep. Ocasio-Cortez: pause new or expanded AI datacenters until Congress writes national rules ([S.4214](#)) · introduced; long shot ([AP](#))
- **Ratepayer Protection Act** — Reps. Gabe Evans + Kathy Castor: big datacenters (100+ MW) cover the cost of power upgrades built for them; House passed 417–3, Senate blocked ([H.R.9340](#))
- **GRID Savings Act** — Sen. Martin Heinrich: federal energy regulators set rules so huge power users pay to plug into the grid ([S.5199](#))
- **Power for the People Act** — Sen. Chris Van Hollen: waiting line for datacenter power hookups, plus a separate rate so their costs stay on their bill ([S.3682](#))



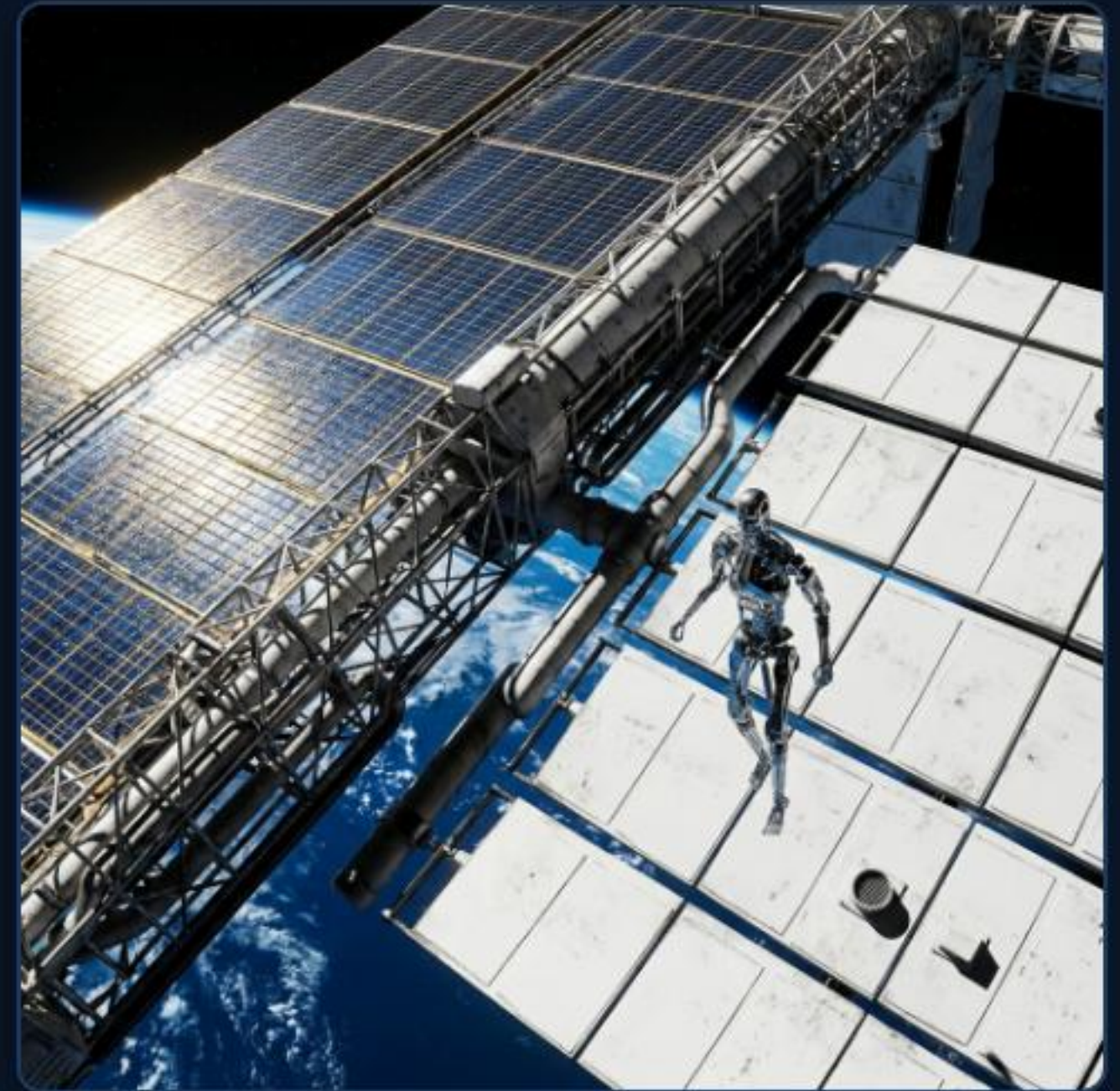
New Datacenters

- **Meta Hyperion** (Louisiana) — up to **5 GW**; >\$50B; building now, full size ~2030–32 ([CNBC](#))
- **OpenAI** (Pike County, Ohio) — about **8 GW** of compute capacity leased; first **800 MW** targeted for 2028 ([OpenAI](#))
- **Microsoft Fairwater** (Wisconsin) — analysts project ~**2+ GW** of IT power by ~2028 as the campus grows ([Epoch AI](#))
- **OpenAI Stargate** (Abilene, TX) — flagship campus about **1.2 GW**; more U.S. sites in the pipeline ([Epoch AI](#))
- For scale: **1 GW $\approx$ 800,000 U.S. homes** (avg ~1.2 kW) — one campus can rival a big city ([EIA](#))



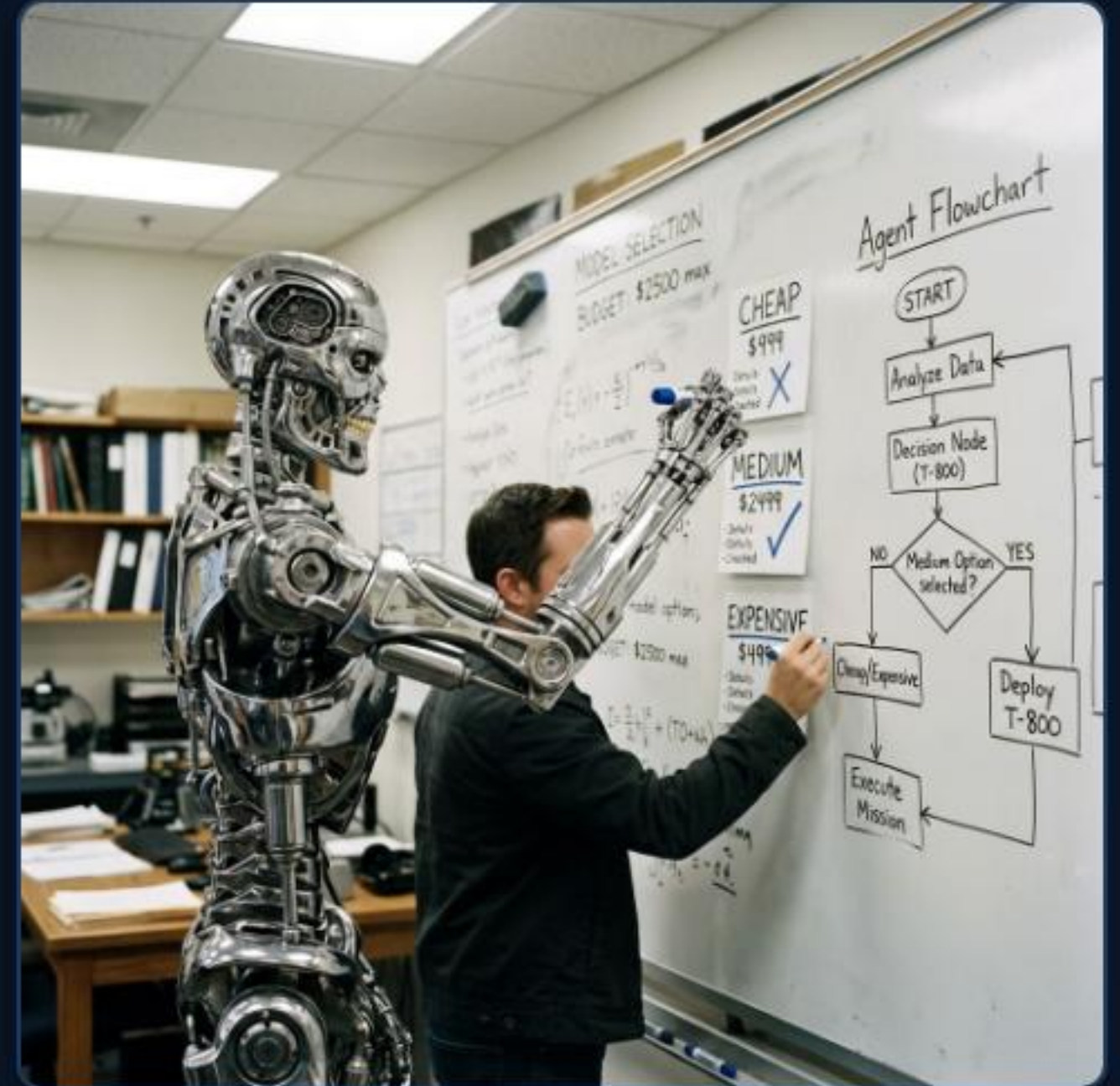
Orbital Datacenters

- **Idea** — compute in orbit: near-constant sun for power; radiators dump heat to space ([Starcloud](#))
- **Starcloud** — NVIDIA-backed startup with AI GPUs already in orbit (H100 chip trained and ran a model in space).
 - Big pitch: multi-GW orbital datacenter with **4 km × 4 km** of solar and cooling panels ([NVIDIA](#))([CNBC](#))
- **SpaceX** — building orbital datacenters ([SpaceX / Yahoo](#))
 - **AI1** — SpaceX's first orbital-datacenter design with power usage of **120 kW**
 - **Starship** — the rocket used by SpaceX, Starcloud, and anyone else, for big launches
- **Engineering challenges**
 - **Radiator mass** — in space you need big metal panels to radiate heat
 - **Launch frequency** — you have to fly all that hardware into space
 - **Latency back to Earth** — lag from laser transit and routing through the fiber-optic network on the ground



Make Agents Cheap

- **Measure** — API responses report token counts; log them
- **Pick cheaper models** — luna for routing; smarter only when needed
- **Caps** — tools, retries, and total tokens
- **Cached tokens** — reuse long prompts (~10× cheaper cached input) ([OpenAI](#))



Cache Token Cost

- Cached tokens let you reuse tokens in input prompt
- Cache long instructions or text you'll reuse (playbooks, books, tool lists)
- Cache write - the first time you write to the tokens to the cache
 - Cache write is bit more expensive than normal input tokens
 - Once written to cache, cached input tokens are MUCH cheaper than normal input tokens

MODEL	INPUT / 1M	CACHED / 1M	CACHE WRITE / 1M	OUTPUT / 1M
gpt-6-luna	\$0.10	\$0.01	\$0.125	\$0.50
gpt-6.1-sol	\$2.00	\$0.10	\$2.50	\$10.00
gpt-6-astra	\$10.00	\$1.00	\$12.50	\$50.00

Live: [OpenAI pricing](#)

Cached Tokens

- Put the long, unchanged stuff **first** in your prompt.
- Use `prompt_cache_breakpoint` to a mark where your cached tokens stop
- Everything after is normal input tokens

lease agent — cache the playbook

Copy

```
PLAYBOOK = open("lease_playbook.txt").read() # long, same every call

def ask(question: str):
    r = client.responses.create(
        model="gpt-6-luna",
        input=[
            {
                "role": "developer",
                "content": [{
                    "type": "input_text",
                    "text": PLAYBOOK, # stable → cache here
                    "prompt_cache_breakpoint": {"mode": "explicit"},
                }],
            },
            {"role": "user", "content": question}, # changes each call
        ],
        prompt_cache_options={"mode": "explicit", "ttl": "30m"},
    )
    d = r.usage.input_tokens_details
    print("cached:", d.cached_tokens, "wrote:", d.cache_write_tokens)
    return r.output_text

ask("Can the tenant sublet?") # 1st call: writes cache (1.25x)
```


Measuring Token Usage

- OpenAI **Responses API** lets you measure token usage with its `usage` field ([docs](#)).
- `input_tokens` — total tokens you sent in (prompt, tools, etc.)
- `output_tokens` — total tokens the model generated (answer + hidden reasoning, if any)
- `total_tokens` — `input_tokens` + `output_tokens`
- `input_tokens_details.cached_tokens` — input read from the prompt cache (cheap reuse)
- `input_tokens_details.cache_write_tokens` — input newly written into the cache this call
- `output_tokens_details.reasoning_tokens` — internal “thinking” tokens on reasoning models (count toward output; often not shown as text)

```
r.usage
├─ input_tokens
├─ input_tokens_details
│   └─ cached_tokens
│       └─ cache_write_tokens
├─ output_tokens
├─ output_tokens_details
│   └─ reasoning_tokens
└─ total_tokens
```

Field names from the OpenAI Python SDK `ResponseUsage` type.

Cheaper Choices

- Newer **choice** models take a rich input (long text, ticket, essay) but output only a pick + **probabilities** — no long essay reply to pay for
- Use this when the agent only needs to make a decision (i.e. route job to another agent)
- **Choice AI models**
 - **Jev** (TypeSafe) — typed choice / score / yes-no with calibrated probs ([Jev](#))
 - **Decisions API** (OpenAI, preview) — Luna-powered picks among predefined answers ([overview](#))

request

Copy

```
{
  "model": "jev-latest",
  "state": "The quick brown fox jumps over the lazy dog.
I wrote this myself in class today.",
  "questions": {
    "origin": {
      "type": "choice",
      "instructions": "Was this text written by a human or by AI?",
      "criteria": {
        "human": "Written by a person",
        "ai": "Generated by an AI model"
      }
    }
  }
}
```

response

Copy

```
{
  "answers": {
    "origin": {
      "type": "choice",
      "choice": "human",
      "probabilities": { "human": 0.82, "ai": 0.18 },
      "confidence": 0.71
    }
  },
  "usage": { "input_tokens": 48, "output_tokens": 0 }
}
```



| Track The Token Bill

- Governments and AI companies: chips, datacenters, power, water
- You: tokens
- Vibe: add a **token tracker** to the Harry Potter multi-agent dashboard
- [Vibe coding session](#) →

